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Can OCR Rules Replace Azure OpenAI?

Medical receipt extraction benchmark

9 receipts · 6 fields · rules vs AI extraction

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Executive Summary

OCR + rules is not enough to replace AOAI across varied receipts.

96.1%

Azure DI + AOAI (prompt v2)

Development result

51.0–70.6%

OCR + general multilingual rules

51.0% = PaddleOCR Light + rules · 70.6% = Azure DI + rules

Why AOAI may still be needed

- Fewer rules to write and maintain for new layouts and scenarios
- Can potentially extract more fields for analytics (not yet benchmarked)
- Rules remain fine for simple, clearly labelled checks

Caveat: 9 receipts. 96.1% is a development result (prompt refined on these receipts); 51.0–70.6% is a retrospective baseline. Neither is independent validation.

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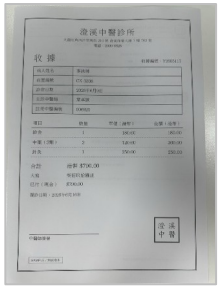
What We Compared

For each of the 9 receipts, we extract 6 key fields and compare the results across 3 approaches.

	1. Read the text (OCR)		2. Find the fields		
PaddleOCR Light + rules	**PaddleOCR Light** open source, own compute	→	**Hand-written rules**	→	**6 fields**
Azure DI + rules	**Azure Document Intelligence** cloud OCR	→	**Hand-written rules**	→	**6 fields**
Azure DI + AOAI	**Azure Document Intelligence** cloud OCR	→	**Azure OpenAI instructions**	→	**6 fields**


English


Bilingual


Chinese

9 receipts: 7 printed test receipts (English, bilingual, Chinese layouts) and 2 from the first benchmark round.

6 fields: doctor, registration no., patient, receipt no., service date, total.

51 of 54 possible fields were scored. 3 receipts do not contain a doctor registration number, so those fields were excluded.

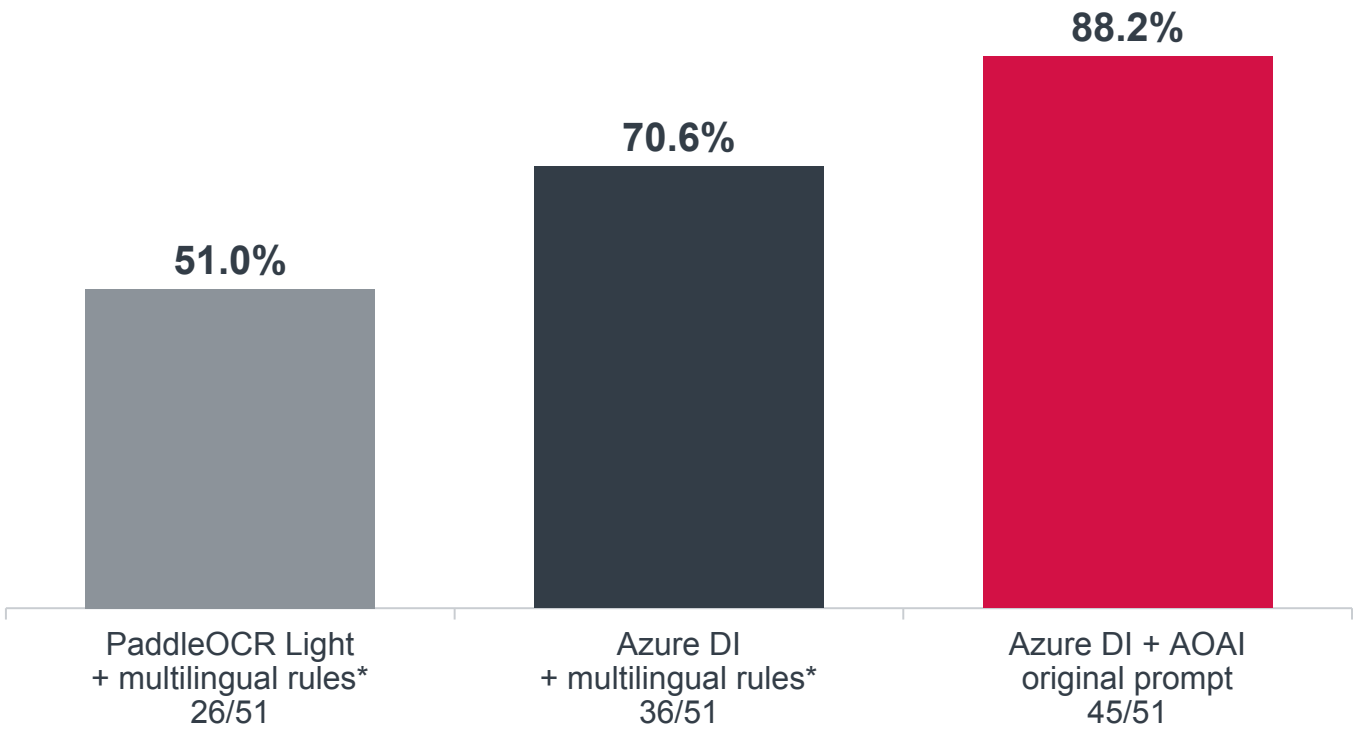
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A Fairer Comparison: Rules vs AOAI

MEASURED · RETROSPECTIVE

With multilingual rules, OCR + rules reached 51.0–70.6%. AOAI reached 88.2% before any prompt tuning.



Multilingual rules*

- General rules plus common Chinese labels and basic bilingual handling
- No receipt-specific tuning

To close the gap, rules would need

- New label types and layouts written by hand
- OCR-tolerant matching and retesting of every receipt

** Retrospective: Chinese labels added after these receipts were examined (rules fixed before scoring); not an independent test. Accuracy = correct fields out of 51 on 9 receipts.*

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The Key Difference: Maintenance

MEASURED · DEVELOPMENT SET

Rules need us to anticipate every case in code. AOAI can often cope through plain-language instructions.

Manual extraction rules

A new label, language or layout appears → A developer anticipates it and writes rule code → Retest every receipt; a change can break a working one

Seen in this benchmark

- 69 → 225 label phrases written by hand
- 1 correct field broken by a rule change

AOAI extraction

New wording appears → Often handled by the existing instructions → If not: refine the instructions in plain language, rerun

Seen in this benchmark

- One prompt revision: 4 fields fixed, 0 broken

Not guaranteed: AOAI has not been tested on enough unseen layouts. It still makes mistakes and needs testing and monitoring.

Where rules were enough: a yes/no check for a diagnosis comment under clear labels. Rules matched AOAI on 13/13 receipts (development result).

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Advantages and Disadvantages

Rules win on cost, auditability and predictability; AOAI on accuracy, maintenance and flexibility.

	OCR + rules	Azure DI + AOAI	Evidence
Cost	+ PaddleOCR: own compute, no API fee (Azure DI: page fee)	− Azure DI page fee + ~602 input / 263 output tokens per receipt	*Prices to confirm*
Speed	PaddleOCR 13.9 s (CPU) · Azure DI 3.8 s	6.2 s per receipt	Measured
Accuracy	− 51.0–70.6% with multilingual rules (retrospective)	+ 88.2% first; 96.1% with prompt v2 (development)	Measured
Maintenance	− New labels, languages and layouts need code and a full retest	+ Many changes made in plain-language instructions; still retest	*Measured / expected*
Flexibility	− Finds only what it was written for	+ New fields can be requested in the prompt (not yet benchmarked)	*Expected*
Auditability	+ Every result traceable to a rule	− Harder to explain; keep quoted evidence and logs	*Expected*
Reliability	+ Same input, same output	− Can return a confident wrong value (e.g. a sub total as the total); model updates need retesting	*Measured / expected*

Green = stronger side. Multilingual rules include common Chinese labels (retrospective baseline). Expected = not measured here.

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Beyond Six Fields: What AOAI Could Unlock

POTENTIAL · NOT YET TESTED

Our receipts already carry medical details. AOAI could turn them into data for analytics.

Type	Text already in our OCR	Receipt
Medication	Omeprazole 20mg Capsules	Pharmacy
Medication	Metronidazole 50mg Tablet	Vet invoice
Test	Chest X-ray	Hospital
Treatment	針灸 (acupuncture)	Chinese medicine
Treatment	Scaling and polishing	Dental
Symptom	diarrhoea	Vet invoice
Diagnosis	Colitis	Vet invoice

Which could enable

- Claims analytics by diagnosis, treatment and medication
- Cost benchmarks per treatment and provider
- Medication and treatment patterns over time
- Consistency checks: does the treatment fit the diagnosis and amount?
- Faster triage and fewer manual look-ups for reviewers

Potential, not yet benchmarked. Text shown is OCR input, not AOAI output. A pilot needs ground truth and a privacy review.

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Limits of This Evidence

The direction is clear, but the numbers are not yet proof.

1

Small dataset

9 receipts, 51 fields. 7 are printed test receipts.

2

Retrospective rules baseline

Multilingual rules (51.0–70.6%) were written after these receipts were examined.

3

Tuned on the same receipts

Prompt v2's 96.1% is a development result. Tuned rule scores are not shown.

4

Little independent validation

Only 4 unseen handwritten receipts. On all 4, Azure DI misread the year 2026 as 2020.

5

Costs not confirmed

Token counts are measured; Document Intelligence and gpt-5-mini prices are not.

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Recommendation

Use rules for simple, labelled checks. Use AOAI where receipts vary or richer data is needed.

Simple OCR rules are sufficient when…

- Layouts are few, known and stable
- Fields have clear printed labels
- The question is simple, e.g. is a diagnosis comment present?
- No API fee and fully traceable output are required

AOAI adds real value when…

- Providers, layouts and languages vary (Chinese, bilingual)
- Fields lack consistent labels: names with titles, totals vs subtotals
- We want less scenario-specific code to maintain
- Richer medical data is wanted (potential, not yet benchmarked)

1. Freeze

the rules and AOAI prompt v2

2. Validate

30+ unseen, real, anonymised receipts; count rule changes

3. Pilot extra fields

medications, treatments, symptoms, diagnoses with ground truth

4. Confirm cost

Document Intelligence pages and gpt-5-mini tokens

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